Shahab Sherafat

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Education

2023 – 2028 **B.Sc. Computer Engineering** at Department of ECE, University of Tehran, Tehran, Iran
Cumulative GPA: 19.20/20.00 (Top 8% of class); **Major GPA**: 19.07/20.00
Minor in Psychology (GPA: 18.50/20.0)

2020 – 2023 **High School Diploma**, Sadat School, Isfahan, Iran
GPA: 19.70/20.00

Research Interests

- Robustness
- Cognitive Science
- Generative AI
- Explainable AI
- Computational Neuroscience
- AI for Medicine

Research Experience

Research Assistant *2026–Present*

– Helmholtz Munich (Institute of AI for Health)
Uncertainty Quantification in Neural Cellular Automata (NCA)
Developing uncertainty quantification frameworks for Neural Cellular Automata (NCA) in biomedical image segmentation. Evaluating model reliability and predictive confidence on benchmark cytological datasets.
Supervisors: Prof. Carsten Marr and Dr. Ario Sadafi

Research Assistant *2026–Present*

– Sharif University of Technology (RIML Lab)
Adversarial Robustness and Transferability in Speech Foundation Models
Developing and evaluating methods to improve adversarial robustness in speech foundation models and transfer robustness to specialized downstream models. Associated manuscripts:
Robustness after Specialization: Installing Adversarial Robustness in Speech Foundation Model Descendants — submitted to *ICLR*.
Transferable Adversarial Robustness for Speech Foundation Models via Hierarchical Stabilization — submitted to *ICASSP*.
Supervisor: Prof. Mohammad Hossein Rohban

Research Assistant *2025–2026*

– University of Tehran (AI for Medicine Lab)
Generative Modeling for MRI-Based Classification
Evaluated generative deep learning models (Diffusion, VAE, LDM) for MRI-based disease classification, using self-supervised representations to mitigate clinical data scarcity.
Supervisors: Prof. Ahmad Kalhor and Dr. Saeed Masoudnia

Research Assistant *2025*

– TEIAS (Tehran Institute for Advanced Studies)
LLM-Based Support for Undergraduate Mathematics Learning
Designed an LLM-based chatbot and agent for a randomized controlled study investigating the effects of LLMs on undergraduate mathematics learning. Evaluated learning retention and automated feedback. Co-authored a paper accepted at *ICCE 2026* [arXiv].
Supervisor: Prof. Behnam Bahrak

Teaching Experience

Teaching Assistant Sharif University of Technology, Department of Computer Engineering

- **Engineering Probability and Statistics** (Head of Final Project Team) *2026–Present*
  Prof. Motahari

Teaching Assistant University of Tehran, Department of Electrical and Computer Engineering

- **Machine Learning** *2026–Present*
  Prof. Tavassolipour, Prof. Abolghasemi
- **Engineering Probability and Statistics** *2026–Present*
  Prof. Haghi
- **Artificial Intelligence** *2025–Present*
  Prof. Yaghoobzadeh, Prof. Fadaei
- **Engineering Probability and Statistics** *2025–2026*
  Prof. Bahrak, Prof. Vahabie
- **Introduction to Computing Systems and C Programming** *2024–2026*
  Prof. Hashemi, Prof. Moradi

Relevant Courses

- **Artificial Intelligence** **Grade: 20/20**
- **Engineering Probability and Statistics** **Grade: 18.7/20**
- **Data Science** **Grade: 20/20**
- **Advanced Programming** **Grade: 20/20**
- **Engineering Mathematics** **Grade: 19.7/20**
- **Design and Analysis of Algorithms** **Grade: 19/20**

Certificates and Honors

Ranked 78th in the National University Entrance Exam 2023
Machine Learning Specialization Andrew Ng Course, Coursera 2025
ACM SOC in Deep Learning, University of Tehran 2024

Extracurricular & Volunteer Experience

- **Official Member of the ACM Student Chapter**, University of Tehran 2025–Present
  Participating in academic and technical activities organized by the association, managing and holding technical and non-technical events contributing to the student tech community.

Skills

Programming Languages	Python, C/C++, R, Java, Bash, SQL, Verilog HDL
Frameworks & Libraries	PyTorch, Hugging Face (Transformers), TensorFlow, Scikit-Learn, NumPy, Pandas, Matplotlib
Tools & Environments	Git, Docker, Linux/WSL
Languages	Persian (Native), English (Proficient)